Lecture 1

Groups acting

Objectives. *The course asks what the structure of a finite group is, and how its linear representations reflect it. This lecture defines groups and sets up the basic tool of the course, a group* acting *on a set. From its main result, orbit–stabiliser, follow Lagrange's theorem, the class equation, orbit counting and Cauchy's theorem.*

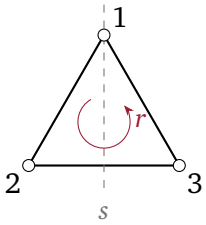
By the end of this lecture you will be able to:

- Check that a rule is an action, and find its orbits, stabilisers and kernel.
- Prove orbit–stabiliser, and deduce Lagrange's theorem and the class equation.
- Find the classes and centralisers of a small group, and write its class equation.
- Count orbits by averaging fixed points; prove Cauchy's theorem.

1.1 Groups, subgroups, homomorphisms

This course studies finite groups, the algebra of symmetry. We begin with an example that shows what a group is, the symmetries of an equilateral triangle.

A *symmetry* of the triangle is an orthogonal map of the plane that carries it to itself. The triangle is centred at the origin, with vertices numbered anticlockwise:



The arrow shows the rotation r by $2\pi/3$, which moves the vertices $1 \mapsto 2 \mapsto 3 \mapsto 1$. The dashed line is the axis of the reflection s, which fixes vertex 1 and exchanges vertices 2 and 3.

Symmetries can be composed. The product rs means first s, then r, and the result is again a symmetry. Composition of maps is associative. The identity map is a symmetry, and every symmetry can be undone by its inverse map, which is also a symmetry. A group is any set with a product that has these properties.

Definition 1.1 (Group). A *group* is a set G with a product $G \times G \to G$, $(g,h) \mapsto gh$, such that

- $(gh)k = g(hk)$ for all $g, h, k \in G$;
- there is an *identity* $e \in G$, with $eg = ge = g$ for all $g \in G$;
- each $g \in G$ has an *inverse* $g^{-1} \in G$, with $gg^{-1} = g^{-1}g = e$.

The *order* $|G|$ of G is its number of elements. The group G is *abelian* if $gh = hg$ for all $g, h \in G$.

The definition asks for an identity and for an inverse of each element. We first check that each of them is unique, so that the symbols e and g^{-1} name definite elements. If e and e' are identities, then

$$e = ee' = e',$$

first because e' is an identity and then because e is. If h and h' are inverses of g, then

$$h = h(gh') = (hg)h' = h',$$

using $gh' = e$, associativity and $hg = e$.

To undo a product, we undo its factors in the reverse order: $(gh)^{-1} = h^{-1}g^{-1}$, since $(gh)(h^{-1}g^{-1}) = g(hh^{-1})g^{-1} = e$, and likewise in the other order. We write g^k for the product of $k \geq 1$ copies of g, and $g^0 = e$.

The main examples of this course are the groups of all bijections of a set.

Definition 1.2 (Symmetric group)**.** The *symmetric group* S_X of a set X is the set of all bijections $X \to X$, with composition as the product. For $X = \{1, \ldots, n\}$ we write S_n and call its elements *permutations*.

Composition is associative, the identity map id is the identity of S_X, and the inverse map of a bijection is its inverse, so S_X is a group. The group S_n has order $n!$.

A *cycle* $(a_1\, a_2\, \ldots\, a_k)$, with distinct entries, is the permutation that sends $a_1 \mapsto a_2 \mapsto \cdots \mapsto a_k \mapsto a_1$ and fixes every other point. It is also called a *k-cycle*, and a 2-cycle is a *transposition*. Two cycles are *disjoint* if they have no entry in common. A one-cycle (i) fixes i, and we usually omit one-cycles when writing a permutation. For example, in S_3 the expressions $(1\,2)(3)$ and $(1\,2)$ denote the same permutation.

We write permutations as products of disjoint cycles, which is always possible (Proposition 2.13). Permutations compose as functions, right to left: $(\sigma\tau)(i) = \sigma(\tau(i))$. For example,

$$(1\,2)(1\,3) = (1\,3\,2).$$

Indeed, applying the right factor first gives $1 \mapsto 3 \mapsto 3$, $3 \mapsto 1 \mapsto 2$ and $2 \mapsto 2 \mapsto 1$.

Example 1.3 (Cyclic and dihedral groups)**.**

- Let $n \geq 1$ be an integer. The nth roots of unity are the complex numbers z with $z^n = 1$. They form a group under multiplication:

  $$C_n = \{\exp(2\pi \mathrm{i}k/n) : k = 0, \ldots, n-1\}.$$

  Every element of C_n is a power of $\exp(2\pi\mathrm{i}/n)$. For this reason C_n is called the *cyclic group* of order n. Definition 1.16 defines the word "cyclic" for general groups.
- The *dihedral group* D_n ($n \geq 3$) is the group of orthogonal maps of $\mathbb{R}^2$ that carry a regular n-gon to itself. The n-gon is centred at 0, with vertices numbered $1, \ldots, n$ anticlockwise. For $n = 3$ these maps are the symmetries of the triangle above. Let r be the rotation by $2\pi/n$ and s the reflection in the line through vertex 1. A symmetry keeps the distance from 0, and the vertices are the points of the n-gon farthest from 0, so a symmetry maps vertices to vertices. We have

  $$r^n = e, \qquad s^2 = e, \qquad srs^{-1} = r^{-1}.$$

  The last relation says that a reflection reverses the sense of a rotation. To check it, read vertex numbers modulo n: two integers name the same vertex when their difference is

divisible by n. Then r sends vertex k to vertex $k+1$, and s sends vertex k to vertex $2-k$. So srs^{-1} sends vertex k to vertex $2-(3-k) = k-1$, as r^{-1} does. A linear map is determined by its values on vertices 1 and 2, so $srs^{-1} = r^{-1}$. Example 1.18 shows that $|D_n| = 2n$.

We now relate groups to each other. A subgroup is a group inside another group, and a homomorphism is a map between groups that respects the products.

Definition 1.4 (Subgroup). A *subgroup* $H \leq G$ is a subset of G that contains e and is closed under products and inverses. It is *proper*, written $H < G$, if $H \neq G$. The *trivial subgroup* is $\mathbf{1} = \{e\}$. The subgroup *generated* by a subset $S \subseteq G$, written $\langle S \rangle$, is the smallest subgroup containing S. We write $\langle g \rangle$ for $\langle \{g\} \rangle$.

The subgroup $\langle S \rangle$ exists: it is the intersection of all the subgroups containing S. For example, in D_3 we have $\langle r \rangle = \{e, r, r^2\}$ and $\langle s \rangle = \{e, s\}$.

Definition 1.5 (Homomorphism). A *homomorphism* from G to H is a map $\varphi : G \to H$ with $\varphi(gh) = \varphi(g)\varphi(h)$ for all $g, h \in G$. Its *kernel* is $\ker \varphi = \{g \in G : \varphi(g) = e\}$, with e the identity of H, and its *image* is $\operatorname{im} \varphi = \{\varphi(g) : g \in G\}$. A bijective homomorphism is an *isomorphism*. If one exists, then G and H are *isomorphic*, written $G \cong H$.

Isomorphic groups differ only in the names of their elements. A homomorphism φ respects identities and inverses. Multiplying $\varphi(e) = \varphi(e)\varphi(e)$ by $\varphi(e)^{-1}$ gives $\varphi(e) = e$, and then $\varphi(g)\varphi(g^{-1}) = \varphi(e) = e$ gives $\varphi(g^{-1}) = \varphi(g)^{-1}$. Hence $\ker \varphi$ and $\operatorname{im} \varphi$ are subgroups. Moreover, $\varphi(g) = \varphi(h)$ if and only if $\varphi(g^{-1}h) = e$, that is, $g^{-1}h \in \ker \varphi$. So φ is injective exactly when $\ker \varphi = \mathbf{1}$.

Example 1.6 ($D_3 \cong S_3$). Each $g \in D_3$ permutes the vertices, and the permutation of gh is that of g after that of h. This gives a homomorphism $D_3 \to S_3$. It is injective, since a linear map fixing vertices 1 and 2, two independent vectors, is the identity. The figure gives the permutations:

g	e	r	r^2	s	rs	r^2s
permutation	e	$(1\,2\,3)$	$(1\,3\,2)$	$(2\,3)$	$(1\,2)$	$(1\,3)$

For instance, rs applies s first, so it sends $1 \mapsto 1 \mapsto 2$, $2 \mapsto 3 \mapsto 1$ and $3 \mapsto 2 \mapsto 3$. The table contains all six elements of S_3, so $D_3 \cong S_3$. This group is the running example of the course.

1.2 Actions

In Example 1.6 the elements of D_3 move the vertices of the triangle. An action describes this situation for any group and any set. Each group element moves the points of the set, and a product moves them as its factors do, one after the other.

Definition 1.7 (Action). A (left) *action* of G on a set X is a map $G \times X \to X$, $(g, x) \mapsto g \cdot x$, with

$$e \cdot x = x,$$
$$g \cdot (h \cdot x) = (gh) \cdot x$$

for all $g, h \in G$ and $x \in X$. The set X is then a *G-set*.

Two sets describe what the group does to a point: the points it can be moved to, and the group elements that leave it in place. We say that g *fixes* x if $g \cdot x = x$.

Definition 1.8 (Orbit and stabiliser)**.** Let G act on X, and let $x \in X$. The *orbit* of x is $G \cdot x = \{g \cdot x : g \in G\}$. The *stabiliser* of x is $G_x = \{g \in G : g \cdot x = x\}$. The action is *transitive* if there is one orbit.

Example 1.9 (D_3 on the vertices)**.** The group D_3 acts on the set $\{1, 2, 3\}$ of vertices: $g \cdot i$ is the image of vertex i under g. The rotations r and r^2 move vertex 1 to vertices 2 and 3, so the orbit of vertex 1 is $\{1, 2, 3\}$ and the action is transitive. The table of Example 1.6 shows that only e and s fix vertex 1, so its stabiliser is $\{e, s\}$.

The subgroup $\langle s \rangle$ also acts on the vertices. Its orbits are $\{1\}$ and $\{2, 3\}$, so this action is not transitive.

The stabiliser G_x is a subgroup of G. It contains e, since $e \cdot x = x$. It is closed under products, since $g \cdot (h \cdot x) = (gh) \cdot x$. It is closed under inverses, because $g \cdot x = x$ gives $x = (g^{-1}g) \cdot x = g^{-1} \cdot (g \cdot x) = g^{-1} \cdot x$.

The orbits partition X. Indeed, "$y = g \cdot x$ for some $g \in G$" is an equivalence relation on X, whose classes are the orbits. It is reflexive, as $x = e \cdot x$. It is symmetric, as $y = g \cdot x$ gives $x = g^{-1} \cdot y$. Finally, the relation is transitive, as $y = g \cdot x$ and $z = h \cdot y$ give $z = (hg) \cdot x$. This is the usual word for relations, and it is unrelated to a transitive action.

Recall that S_X is the group of all bijections $X \to X$ (Definition 1.2). Each element of G moves the points of X by such a bijection, so an action is the same as a homomorphism to S_X.

Proposition 1.10 (Actions are homomorphisms to S_X)**.** *An action of G on X gives a homomorphism $\rho\colon G \to S_X$, where $\rho(g)$ is the bijection $x \mapsto g \cdot x$. Conversely, a homomorphism $\rho\colon G \to S_X$ gives the action $g \cdot x = \rho(g)(x)$. The* kernel *of the action is* $\ker \rho = \bigcap_{x \in X} G_x$.

Proof. The axioms say exactly $\rho(e) = \mathrm{id}$ and $\rho(g)\rho(h) = \rho(gh)$. So $\rho(g^{-1})$ inverts $\rho(g)$, and each $\rho(g)$ is a bijection. Conversely, a homomorphism satisfies these two equations, and they are the axioms. Finally, $\rho(g) = \mathrm{id}$ exactly when g fixes every $x \in X$. ∎

The action is *faithful* if its kernel is $\mathbf{1}$, that is, if only e fixes every point. Then ρ is injective and $G \cong \operatorname{im} \rho \leq S_X$, as for D_3 on the vertices (Example 1.6).

The next action comes from a subgroup, and it is used throughout the course. The set acted on consists of the following subsets of G.

Definition 1.11 (Cosets and index)**.** Let $H \leq G$. The *left coset* of $x \in G$ is $xH = \{xh : h \in H\}$. The set of left cosets is G/H, and its size is the *index* $[G : H]$ of H in G.

Different elements can give the same coset: $xH = yH$ exactly when $x^{-1}y \in H$. If $xH = yH$, then $y \in xH$, so $x^{-1}y \in H$. Conversely, if $y = xh$ with $h \in H$, then $hH = H$, because H is closed under products and inverses, and

$$yH = x(hH) = xH.$$

For example, $H = \{e, (1\,2)\}$ has three left cosets in S_3, so $[S_3 : H] = 3$:

$$H, \qquad (1\,3)H = \{(1\,3), (1\,2\,3)\}, \qquad (1\,3\,2)H = \{(1\,3\,2), (2\,3)\}.$$

Example 1.12 (The action on cosets)**.** For $H \leq G$, the group G acts on G/H by $g \cdot xH = gxH$. This rule is well defined: gxH is the image of the set xH under $y \mapsto gy$, so it depends only on xH. The axioms hold because they hold for the product of G. The action is transitive, since $(yx^{-1}) \cdot xH = yH$.

Example 1.13 (Guard: G/H is a G-set, not a group)**.** The rule $xH \cdot yH = xyH$ does not define a product in general, because its result can change when x is replaced by another element of xH. In S_3 take $H = \{e, (1\,2)\}$ again. Then $eH = (1\,2)H$, but with $y = (1\,3)$ the rule gives two different cosets:

$$\begin{aligned} e\,yH &= \{(1\,3),\ (1\,2\,3)\}, \\ (1\,2)\,yH &= (1\,3\,2)H \\ &= \{(1\,3\,2),\ (2\,3)\}. \end{aligned}$$

Lecture 2 finds the subgroups H for which the rule is well defined, and then G/H is a group. Lecture 8 uses G/H for every H, and needs only the action.

1.3 Orbit–stabiliser and Lagrange

In Example 1.9 the orbit of vertex 1 has 3 elements, its stabiliser has 2, and $3 \cdot 2 = |D_3|$. We now explain this relation. Two elements of G send x to the same point exactly when they lie in the same left coset of G_x. So the points of the orbit of x correspond to these cosets.

Theorem 1.14 (Orbit–stabiliser)**.** *Let G act on X and let $x \in X$. The map*

$$G/G_x \to G \cdot x, \qquad gG_x \mapsto g \cdot x,$$

is well defined and bijective, and it carries the action of G on cosets to the action on the orbit. Hence $|G \cdot x| = [G : G_x]$.

Proof. For $g, h \in G$,

$$\begin{aligned} gG_x = hG_x &\iff g^{-1}h \in G_x \\ &\iff g^{-1}h \cdot x = x \\ &\iff h \cdot x = g \cdot x. \end{aligned}$$

Read from left to right, the chain says the map is well defined; read from right to left, that it is injective. It is onto $G \cdot x$ by the definition of the orbit. Finally $k \cdot gG_x = kgG_x$ goes to $kg \cdot x = k \cdot (g \cdot x)$. ∎

Lagrange's theorem follows by letting a subgroup act on the whole group.

Corollary 1.15 (Lagrange)**.** *If G is finite and $H \leq G$, then $|G| = [G : H]\ |H|$. Hence $|H|$ divides $|G|$. Moreover, for any action of G on a set X and any $x \in X$, $|G \cdot x| = |G|\,/|G_x|$.*

Proof. *The action.* Let H act on G by $h \cdot g = gh^{-1}$. The identity fixes every g, and

$$h \cdot (h' \cdot g) = gh'^{-1}h^{-1} = g(hh')^{-1} = (hh') \cdot g.$$

The orbits. The orbit of g is gH. Its stabiliser is $\mathbf{1}$, since $gh^{-1} = g$ forces $h = e$. By Theorem 1.14, each orbit has $[H : \mathbf{1}] = |H|$ elements.

The count. The orbits partition G into its $[G : H]$ left cosets, so $|G| = [G : H]\ |H|$. Apply this formula with $H = G_x$, then use orbit–stabiliser, to obtain the final claim. ∎

We apply this to the subgroup formed by the powers of one element. In a finite group the powers of g repeat, and $g^i = g^j$ with $i < j$ gives $g^{j-i} = e$. So some positive power of g is e.

Definition 1.16 (Order of an element, cyclic group)**.** The *order* $o(g)$ of an element g of a finite group is the least $k \geq 1$ with $g^k = e$. A group G is *cyclic* if $G = \langle g \rangle$ for some $g \in G$.

The same argument shows that $e, g, \dots, g^{o(g)-1}$ are distinct. For $m \geq 0$, write $m = q\,o(g)+t$ with $0 \leq t < o(g)$. Then $g^m = g^t$, so $g^m = e$ exactly when $o(g)$ divides m. Hence these $o(g)$ powers are closed under products. They contain the inverse $g^{o(g)-i}$ of g^i, so they form $\langle g \rangle$. So $|\langle g \rangle| = o(g)$, and the two uses of the word "order" agree.

Now let $G = \langle g \rangle$ have order n, so that $o(g) = n$. For $0 \leq j \leq k$, we have $g^j = g^k$ exactly when $g^{k-j} = e$, that is, when n divides $k - j$, that is, when $\exp(2\pi \mathrm{i} j/n) = \exp(2\pi \mathrm{i} k/n)$. So $g^k \mapsto \exp(2\pi \mathrm{i} k/n)$ is a well-defined injective map $G \to C_n$, and it is onto since both sets have n elements. It sends $g^j g^k = g^{j+k}$ to the product of the images of g^j and g^k, so it is an isomorphism. The group C_n is itself cyclic, generated by $\exp(2\pi \mathrm{i}/n)$. So C_n is, up to isomorphism, the cyclic group of order n.

Corollary 1.17. *In a finite group, $o(g)$ divides $|G|$. A group of prime order p is cyclic, generated by any $g \neq e$.*

Proof. Apply Corollary 1.15 to $\langle g \rangle$. If $|G| = p$ and $g \neq e$, then $o(g)$ divides p and is not 1, so $\langle g \rangle = G$. ∎

Example 1.18 ($|D_n| = 2n$)**.** The group D_n acts on the n vertices of the polygon, since a symmetry maps vertices to vertices. The group is finite: a symmetry is linear, so it is determined by the images of the independent vectors at vertices 1 and 2, and these images are vertices. So D_n has at most n^2 elements. The action is transitive, since r^k takes vertex 1 to vertex $k + 1$.

We find the stabiliser of vertex 1. A symmetry that fixes vertex 1 maps vertices to vertices and keeps distances, so it sends vertex 2 to a neighbour of vertex 1, which is vertex 2 or vertex n. Its images of vertices 1 and 2 determine it, so there are at most two such symmetries, and e and s are two of them. The stabiliser is therefore $\{e, s\}$, and Corollary 1.15 gives $|D_n| = 2n$.

The $2n$ elements r^k and $r^k s$ ($0 \leq k < n$) are distinct, because $o(r) = n$ and $\det r^k s = -1$. So

$$D_n = \{e, r, \dots, r^{n-1}, s, rs, \dots, r^{n-1}s\}.$$

1.4 Conjugation and the class equation

A group also acts on itself, by *conjugation*. We write ${}^g x = gxg^{-1}$, and call this element a *conjugate* of x. This is an action, since the identity fixes x and

$${}^g({}^h x) = ghxh^{-1}g^{-1} = {}^{gh}x.$$

For example, in D_n the conjugate $srs^{-1} = r^{-1}$ of r is a rotation through the same angle in the opposite sense. The orbits and stabilisers of conjugation have their own names.

Definition 1.19 (Classes, centralisers, centre)**.** The orbit of g under conjugation is its *conjugacy class* $[g] = \{hgh^{-1} : h \in G\}$. The stabiliser of g is its *centraliser* $C_G(g) = \{h \in G : hg = gh\}$. The *centre* of G is $Z(G) = \{z \in G : zh = hz \text{ for all } h \in G\}$.

The centraliser consists of the elements that commute with g, since $hgh^{-1} = g$ means $hg = gh$. An element z lies in $Z(G)$ exactly when $[z] = \{z\}$. The group G is abelian exactly when every class has one element. The centre $Z(G)$ is also the kernel of conjugation, which is therefore faithful exactly when $Z(G) = \mathbf{1}$.

Orbit–stabiliser now gives the size of each class. Since the classes partition G, adding their sizes gives an equation for $|G|$.

Theorem 1.20 (The class equation)**.** *Let G be finite.*

(i) $|[g]| = [G : C_G(g)]$*, which divides* $|G|$*.*

(ii) Let $g_1, \dots, g_m$ *be one element from each class with more than one element. Then*

$$|G| = |Z(G)| + \sum_{i=1}^{m} [G : C_G(g_i)],$$

and each term of the sum divides $|G|$ *and exceeds* 1*.*

Proof. (i) This is Theorem 1.14 for conjugation, with Corollary 1.15.

(ii) The classes partition G. The classes with one element are the $\{z\}$ with $z \in Z(G)$; the others have $[G : C_G(g_i)] > 1$ elements, by the class-size formula. ∎

Example 1.21 (The classes of S_3)**.** We find each class from the order of a centraliser. First,

$$(1\,3)(1\,2)(1\,3)^{-1} = (2\,3),$$
$$(2\,3)(1\,2)(2\,3)^{-1} = (1\,3),$$
$$(1\,2)(1\,2\,3)(1\,2)^{-1} = (1\,3\,2).$$

So $C_{S_3}((1\,2))$ is a proper subgroup containing $\{e, (1\,2)\}$. By Corollary 1.15, its order divides 6 and is even, so it is 2. Thus the class of $(1\,2)$ has 3 elements, the three transpositions. Likewise $C_{S_3}((1\,2\,3))$ is proper and contains $\langle(1\,2\,3)\rangle$, of order 3. Thus the class of $(1\,2\,3)$ has 2 elements, the two 3-cycles. With $\{e\}$, these classes hold all six elements:

class	representative g	$\lvert[g]\rvert$	$C_{S_3}(g)$
$\{e\}$	e	1	S_3
$\{(1\,2), (1\,3), (2\,3)\}$	$(1\,2)$	3	$\{e, (1\,2)\}$
$\{(1\,2\,3), (1\,3\,2)\}$	$(1\,2\,3)$	2	$\langle(1\,2\,3)\rangle$

So $Z(S_3) = \mathbf{1}$, and the class equation reads

$$6 = 1 + 3 + 2.$$

Lecture 2 finds the classes of every S_n.

Conjugation moves subgroups too. The map $x \mapsto gxg^{-1}$ is an isomorphism $G \to G$, with inverse $x \mapsto g^{-1}xg$, so for $H \leq G$ the image $gHg^{-1} = \{ghg^{-1} : h \in H\}$ is a subgroup, a *conjugate* of H. Moreover, the rule $g \cdot H = gHg^{-1}$ defines an action of G on the set of its subgroups, by the computation that made conjugation an action.

Corollary 1.22. *For $H \leq G$, the stabiliser of H under this action is its* normaliser $N_G(H) = \{g \in G : gHg^{-1} = H\}$*, and H has $[G : N_G(H)]$ conjugates.*

This is Theorem 1.14 for the action on subgroups.

Example 1.23 (Conjugate subgroups of S_3)**.** The conjugates of $\langle(1\,2)\rangle$ are generated by the conjugates of $(1\,2)$, so they are

$$\langle(1\,2)\rangle, \quad \langle(1\,3)\rangle, \quad \langle(2\,3)\rangle.$$

The normaliser of $\langle(1\,2)\rangle$ therefore has index 3, so it has order 2. It contains $\langle(1\,2)\rangle$, so it equals $\langle(1\,2)\rangle$. In contrast, $\langle(1\,2\,3)\rangle$ contains both 3-cycles, so it is its only conjugate.

Lecture 2 calls a subgroup with $N_G(H) = G$ *normal*. Lecture 3 uses normalisers to count the subgroups of largest prime-power order.

1.5 Cayley and orbit counting

By Proposition 1.10, a faithful action of G on X places G inside S_X. Every group has such an action, namely left multiplication on itself.

Theorem 1.24 (Cayley). *Every group G is isomorphic to a subgroup of S_G. If $|G| = n$, it is isomorphic to a subgroup of S_n.*

Proof. Let G act on itself by left multiplication, $g \cdot x = gx$. Since $g \cdot e = g$, only e fixes e, so the kernel is $\mathbf{1}$ and the action is faithful. By Proposition 1.10, $G \cong \operatorname{im}\rho \leq S_G$. Numbering the elements of G as $1, \dots, n$ identifies S_G with S_n. ∎

So every finite group is a group of permutations. The symmetric group in Cayley's theorem can be much larger than needed: it puts D_3 into S_6, whereas the vertices gave $D_3 \cong S_3$.

Orbit–stabiliser also counts orbits: their number is the average number of points fixed by an element of G. This result is often called Burnside's lemma.

Theorem 1.25 (Orbit counting). *Let a finite group G act on a finite set X, and let $\operatorname{Fix}(g) = \{x \in X : g \cdot x = x\}$. The number of orbits is*

$$\frac{1}{|G|} \sum_{g \in G} |\operatorname{Fix}(g)| .$$

Proof. Count the set $P = \{(g, x) \in G \times X : g \cdot x = x\}$ in two ways. By g, $|P| = \sum_g |\operatorname{Fix}(g)|$. By x, and by Corollary 1.15,

$$|P| = \sum_{x \in X} |G_x| = \sum_{x \in X} \frac{|G|}{|G \cdot x|}.$$

Group the x by orbits: an orbit O contributes $|O|$ terms, each $|G|/|O|$, so $|G|$ in all. Hence $|P|$ is $|G|$ times the number of orbits. ∎

For D_3 on the vertices, e fixes 3 points, each of s, rs and $r^2 s$ fixes 1 by the table of Example 1.6, and r and r^2 fix none. The number of orbits is $(3 + 1 + 1 + 1)/6 = 1$, as in Example 1.9.

1.6 Prime-power groups and Cauchy's theorem

By Lagrange, the order of every element divides $|G|$. Cauchy's theorem is a partial converse: every prime divisor p of $|G|$ is the order of an element. Its proof counts fixed points modulo p. For an action of G on X, let $X^G = \{x \in X : g \cdot x = x \text{ for all } g \in G\}$ be the set of points that every element of G fixes.

For integers m and t, the notation $m \equiv t \pmod{p}$ means that p divides $m - t$. Thus the congruence in the next lemma says that the number of points outside X^G is divisible by p.

Lemma 1.26 (Fixed points modulo p). *Let p be prime, let $|G| = p^a$, and let G act on a finite set X. Then $|X| \equiv |X^G| \pmod{p}$.*

Proof. The orbits partition X. By Corollary 1.15 an orbit has $|G|/|G_x|$ elements, a divisor of p^a, so it has one element or a multiple of p. The orbits with one element are the points of X^G. Adding the orbit sizes gives $|X| = |X^G| +$ (a multiple of p). ∎

Theorem 1.27 (Cauchy). *If G is finite and a prime p divides $|G|$, then G has an element of order p.*

Proof. *The set.* Let X be the set of p-tuples $(g_1, \dots, g_p) \in G^p$ with $g_1 g_2 \cdots g_p = e$. The first $p-1$ entries are arbitrary, and they determine $g_p = (g_1 \cdots g_{p-1})^{-1}$. So $|X| = |G|^{p-1}$, a multiple of p.

The action. A cyclic shift keeps a tuple in X:

$$g_p g_1 \cdots g_{p-1} = g_p (g_1 \cdots g_p) g_p^{-1} = e.$$

So C_p, generated by $\zeta = \exp(2\pi \mathrm{i}/p)$, acts on X, with ζ^k shifting each tuple k places. This is well defined, since a shift by p places changes nothing.

The fixed points. A tuple fixed by ζ has all its entries equal, so X^{C_p} consists of the $(g, \dots, g)$ with $g^p = e$. By Lemma 1.26,

$$\left|X^{C_p}\right| \equiv |X| \equiv 0 \pmod{p}.$$

The element. The set X^{C_p} contains $(e, \dots, e)$, so it has at least $p \geq 2$ elements, and some $g \neq e$ has $g^p = e$. Then $o(g)$ divides p and $o(g) \neq 1$, so $o(g) = p$. ∎

Example 1.28 (Cauchy's count in S_3). For $p = 3$, the set X has $6^2 = 36$ elements. The fixed tuples are (g, g, g) with

$$g \in \{e, (1\,2\,3), (1\,3\,2)\}.$$

The fixed-point congruence reads

$$36 \equiv 3 \equiv 0 \pmod{3}.$$

Lecture 3 goes further: it finds a subgroup, rather than an element, whose order is the largest power of p dividing $|G|$.

1.7 Worked example

We count the colourings of the vertices of a triangle with k colours, where colourings that differ by a symmetry count as one.

A colouring is a map $c\colon \{1, 2, 3\} \to \{1, \dots, k\}$; there are k^3. The symmetries act on them, through $D_3 \cong S_3$ (Example 1.6), by

$$(g \cdot c)(i) = c(g^{-1}(i)).$$

Taking $i = g(j)$ gives $(g \cdot c)(g(j)) = c(j)$, so the colour that c put at vertex j moves to vertex $g(j)$. This is an action, since $e \cdot c = c$ and

$$\begin{aligned}(g \cdot (h \cdot c))(i) &= c(h^{-1} g^{-1}(i)) \\ &= ((gh) \cdot c)(i).\end{aligned}$$

The colourings that count as one are exactly the orbits of this action.

Example 1.29 (Guard: the inverse is needed). The rule $g \cdot c = c \circ g$ gives $g \cdot (h \cdot c) = c \circ h \circ g$, while $(gh) \cdot c = c \circ g \circ h$. For $k \geq 2$ these differ: with $g = (1\,2)$, $h = (1\,3)$ and c giving colour 2 to vertex 1 alone, the first gives colour 2 to vertex 3, the second to vertex 2. Lecture 6 and Lecture 14 need the same inverse.

We count the orbits with Theorem 1.25. A colouring satisfies $g \cdot c = c$ exactly when $c(g(j)) = c(j)$ for every vertex j, that is, when c is constant on each cycle of g, with a fixed point counted as a cycle. So $|\mathrm{Fix}(g)| = k^{(\text{number of cycles of } g)}$:

g	e	transpositions	3-cycles
how many	1	3	2
$\lvert\mathrm{Fix}(g)\rvert$	k^3	k^2	k

Hence the number of orbits is

$$\frac{k^3 + 3k^2 + 2k}{6}.$$

For $k = 2$ the count is

$$\frac{8 + 12 + 4}{6} = 4.$$

With the two colours drawn white and filled, and each triangle placed as in the first figure, the eight colourings fall into four orbits, separated by gaps, of sizes 1, 3, 3, 1:



For $k = 3$ the count is

$$\frac{27 + 27 + 6}{6} = 10.$$

Self-check. Centre a cube at the origin. Its *rotations* are the orthogonal maps of $\mathbb{R}^3$ with determinant 1 that carry the cube to itself. How many are there? (Use orbit–stabiliser on the faces.)

Answer: 24. Let G be this group of rotations. A rotation keeps the distance from 0, so it permutes the eight vertices, the points of the cube farthest from 0. A linear map is determined by its values on three independent vertex vectors, so G is finite.

A quarter-turn can move a given face to any adjacent face, and a half-turn can move it to the opposite face. Hence the action on the 6 faces is transitive.

A rotation preserving a face fixes its centre, so it fixes the axis through the origin and that centre. On the plane perpendicular to this axis it is a planar rotation. To preserve the square face, its angle must be 0, $\pi/2$, π or $3\pi/2$. All four of these rotations preserve the cube, so the stabiliser of the face has order 4. By Corollary 1.15,

$$|G| = 6 \cdot 4 = 24.$$

Summary.

- *The main result:* $gG_x \mapsto g \cdot x$ identifies G/G_x with $G \cdot x$ (Theorem 1.14). It gives Lagrange (Corollary 1.15), the class equation (Theorem 1.20), orbit counting (Theorem 1.25) and Cauchy's theorem (Theorem 1.27).
- *Handed on:* actions as homomorphisms $G \to S_X$, cosets, the order of an element, cyclic groups, classes, centralisers, normalisers, the fixed-point congruence modulo p, and $S_3 \cong D_3$.
- *Next:* G/H is not in general a group (Example 1.13); Lecture 2 makes G/N one when N is normal, and finds the classes of S_n.

References

Dummit and Foote, *Abstract Algebra*: D_n (their D_{2n}), §1.2, pp. 23–26; actions, §1.7, pp. 41–43; orbit–stabiliser, §4.1, Prop. 2, pp. 114–115; Cayley, §4.2, Cor. 4, p. 120; the class equation, §4.3, Prop. 6 and Thm. 7, pp. 123–124. Rotman, *An Introduction to the Theory of Groups*: Lagrange, Thm. 2.11, p. 26; orbit–stabiliser, Thm. 3.19, p. 57; counting orbits, Thm. 3.22, pp. 58–59; Cauchy, with McKay's proof, Thm. 4.2, p. 74. Humphreys, *A Course in Group Theory*, Ch. 10, pp. 89–94.